An Intelligent Educational and Communication Tool for Students with Visual and Auditory Impairments

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***Abstract*—The modern digital world presents very large communication barriers for people with combined visual and auditory impairments, many times including ineffective participation in the processes of education and social interaction. This paper introduces a mobile application aimed at bridging these gaps with a whole new Morse code-based comprehensive platform that assists in communication and learning. Other features will include Morse Typing Window for Morse-based typing, Text to Morse, and Speech to Morse, which will convert the written and spoken language to Morse. It includes Morse for PDF, integrating Morse into the test files to make them more accessible. Using the Article Reader feature, the user will be able to take articles and convert them to Morse for study. An integrated Tutorial will walk one through how to learn Morse code and how to use the app. An NLP-powered word suggestion model increases the efficiency of typing and thus makes communication smooth and quicker. Tested in a school environment, the application received very positive feedback about its role in empowering students with multiple disabilities. Such a tool is a great example of how advanced information processing combined with accessible design can realize the full potential of intelligent assistive technology to support inclusive education and communication of persons with complex accessibility needs.**

Keywords: Assistive Technology, Disabilities, Communication Devices, Visual Impairment, Auditory Impairment, Morse Code, Mobile Applications, Natural Language Processing, Accessibility, Inclusive Education

I. Introduction

With the world having taken up more of an upheld face towards the digital era, communication is key to most social interactions and education. However, people with combined visual and auditory impairments have very limited opportunities to participate in these spheres of life [1]. The traditional communicative continuum practices are not accessible to them, which makes many of those persons isolated and unable even to express their thoughts or learn anything. This gap underlines the immediate need for creative solutions that could grant people with multiple disabilities communication and learning skills [2].

Such challenges are better addressed by Morse code, given that it is a time-tested communication system. The binary nature of the Morse code, comprising short and long signals, can be translated into tactile feedback quite effectively in order to make things accessible for users [3]. Through this approach, we are able to undertake tool design that allows users to communicate independently and access information in an inclusive manner [4].

This paper presents a mobile application intended to provide a platform for visually and auditorily challenged people to learn and use Morse code. The proposed system introduces a Morse Typing Window that allows users to input Morse code through tactile feedback, thus enabling efficient communication with peers and educators. The Text to Morse feature involves the translation of written text into Morse code, fitting for those who make use of text methods of communication [5]. Another feature is Speech to Morse; besides, one can convert spoken words into Morse code to further expand the modes of communication. The Morse for PDF feature integrates Morse code with PDF documents so that users can gain access to the educational materials in a format compatible with their needs. This allows users to have articles read to them, and if one wants, to change those articles to Morse for studying. The app comes with an integrated Tutorial that teaches Morse code to users, as well as ways of using the application effectively. The application uses an NLP-driven word suggestion model that is capable of predicting and suggesting words when users type in Morse code. It enhances the efficiency of typing and at the same time reduces cognitive overload for the users [6]. This in turn enables smoother, and even faster, communication.

This application has been tested in a school setting, where it received positive feedback from students, thereby underlining its ability to empower the disabled. It becomes a model for how technology can serve inclusive education and communication for individual people with complex access challenges by effectively merging advanced information processing with accessible design. This paper attempts to delve into the development and effectiveness of this innovative solution. It also tries to show how the innovation finds application in today's digital environment to determine its potential to create more inclusion within educational and social processes.

II. State of the Art: History and Context

While communication is the axis around which social interactions and educational engagement revolve, those with

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combined visual and auditory impairments go through so many barriers that inhibit these activities. The World Health Organization estimated that currently, more than 1.5 billion people worldwide are suffering from some form or degree of hearing loss and will increase up to 2.5 billion by 2050, of whom over a quarter are said to be suffering from disabling hearing loss that affects the health and quality of life [7]. Moreover, at least 2.2 billion people in the world have either a near or distance vision impairment; out of these, at least 1 billion of the cases are preventable or still remaining unattended [8]. Of these, a large portion are recorded with dual disabilities both visual-auditory impairments experiencing double disadvantages in communicating and receiving information. These are usually associated with isolation and frustration in educational settings because the traditional teaching methods do not accommodate diversities.

A. Challenges of Traditional Communication Methods

Traditional methods of communication for people with visual and auditory impairments have relied heavily on some form of communication involving tactile sensation, such as tactile sign language, Braille, and special communication boards. Tactile sign language requires a reliance on the sense of touch whereby the individual feels what signs another person is making. However, this then becomes a restrictive technique since it entirely relies on the skills of the user relating to sign language and is rather hard to learn, in which case it may become a barrier to proper communication when one is trying to pass information to a person who has not been trained.

Another traditional method of communication widely used is Braille, which is a tactile writing system used by visually impaired people to access information in written form. At the same time, it does not provide the solutions needed for communicational interaction among individuals with both visual and auditory impairments. Furthermore, specialized communication boards, sometimes with raised symbols or textures, utilize some sort of communication, but these often require significant adjustment and familiarity by the communicator and the one receiving the information. These techniques inadequately prepare an individual with dual disabilities for educational and social settings, often proving to be a hindrance in either fully expressing themselves or comprehending the expressions of others.

The gap in communication increases in the case of people affected with dual disabilities, living in a world devoid of inclusive educational environments and resources. Their unique needs make traditional methods of teaching non-accommodative; hence, they always feel isolated and frustrated. On top of that, the gap in technology just makes things worse. While newer models in assistive technology have leveled the playing field for people with singular disabilities, many tools remain inadequate or completely inaccessible for people with impairments in double categories.

B. Current Technologies and Their Limitations

Recent development in assistive technologies has tried to bridge this communication gap for people suffering from dual disabilities, specifically those having combined visual and auditory impairments. Current technologies include:

Tactile sign language devices provide the facility of tactile sign language by converting signs into either vibrations or tactile symbols. However, the effectiveness of these devices is still limited by requiring both users involved to know tactile sign language [9]. Similarly, Braille displays are electronic displays that provide better access to written information by enabling users to read text from other digital devices. Although these are of immense importance to visually impaired divers, the technology stops here and does not address the auditory aspects of communication; hence this is a problem for people with auditory deficiencies [10].

Speech recognition software dictates spoken languages into text to be presented in order for visually impaired users to present information in that type of format. This does not, however, help the users with auditory deficiencies by this technology. Furthermore, it does not have support for haptic interaction. Moreover, speech recognition is sensitive to errors when the environment is noisy, which jeopardizes its overall efficiency [11].

On a similar note, several AR solutions aim to deliver information regarding visual content through haptic output and other senses. Of these, none are personalized or customized for people with dual disabilities. The mere fact that the messages in AR applications have something to do with vision makes it completely unreachable to those people who cannot see and hence shuts more doors to information access [12]. Furthermore, the communication boards using symbols are made up of symbols for words or sets of words. While these would make communication possible, they require a great deal of adaptation and familiarity on the part of both interlocutors. This level of reliance on the users' understanding may make them unsuitable for any quick or dynamic form of communication, thus placing people with dual disabilities at a disadvantage [13].

Technologies proposed so far rarely offer integrated solutions to put together sight and hearing impairments as a rule. Without communicative devices that have been universally designed, there is a barrier created for the full participation of a person with impairments in both vision and hearing in educational and social contexts, which calls for new solutions that support democratic communication and interaction.

In that respect, Morse code becomes helpful for people who cannot see or hear; it grants them independence in communication, giving signals in the form of tactile feedback or vibration. And this access to communication is further extended by various mobile applications that allow typing in Morse code messages, which one can carry out with the help of taps or vibrations on the device. Therefore, this flexibility of Morse code provides an important means to help achieve this communication in different situations.

C. Comparison of Existing Morse Code Applications

Many other Morse code applications, that bring it to the disabled for communication purposes, fail to provide requisite accessibility features that are essential for dual disabilities. For instance, The Morse-Comm app by Sachdeva

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et al. is an example of a communication application using Morse code with input by gestures. While this may seem novel, the highly dependent nature of an exact recognition of the intended gestures by the interface makes it pretty hard and time-consuming for a user with motor impairments, resulting in miscommunication at times [14].

Similarly, Gboard (Google Keyboard) allows Morse code entry, but taking precise time, which is very challenging for users with fine motor control problems: mistakes will most likely happen frequently and can make communication be misunderstood. The application at best has badly adjusted timings, further decreasing the chances of being accessed [15].

While the Samsung Good Vibes app enables communication for individuals with both visual and auditory deficiencies by converting Morse code into text, it lacks dynamic haptic feedback, an essential component for any user relying on this form of real-time navigation and support for communication. This may weaken its effectiveness for individuals relying heavily on touch in navigating themselves through the environment [16].

The application presented in the paper overcomes these deficiencies by incorporating various features explicitly designed for dual-disabled users. Continuous haptic feedback, in this way, allows users to feel every input, hence helping to increase accuracy and decrease dependence on precise motor skills or visual input. Customizable sensitivity for input addition and real-time timing feedback further smoothen Morse code entry, making it more adaptable. This application goes beyond sheer communication to an unparalleled edge in the field of education. By incorporating real-time word suggestions, it is much easier for the users to input Morse, hence it is a platform for inclusive learning. Further, it enhances both the independent learning and structured interaction with teachers, making this application uniquely suited for educational applications within dual-disabled individuals.

III. Proposed Solution: Enhanced Morse Code Application for Dual-Disabled Students

Building upon the limitations identified in existing technologies and applications, our proposed solution introduces a more inclusive Morse code application that specifically aids users who have both visual and auditory impairments. This will not only provide a communication avenue for dual-disabled persons but will also enhance learning by incorporating ease of use, accessibility, and effective learning. Here, our design incorporates real-time feedback, adaptive input customization, and word suggestions in NLP-driven Morse code to make it more intuitive and responsive to individual user needs than any other application currently available.

The next sections describe the main distinctive features that are the reason for our solution, focusing on innovations in tactile feedback, educational tools, and enhanced accuracy in Morse code entry.

A. Morse Typing Window

The home screen of the application, depicted in Figure 1.a, where buttons to all available features link directly to the respective feature window. Besides navigation through buttons, it also offers Morse code-based input for navigating through the application. Further, users can navigate through this application by voice commands, making it more versatile and friendly for the user.

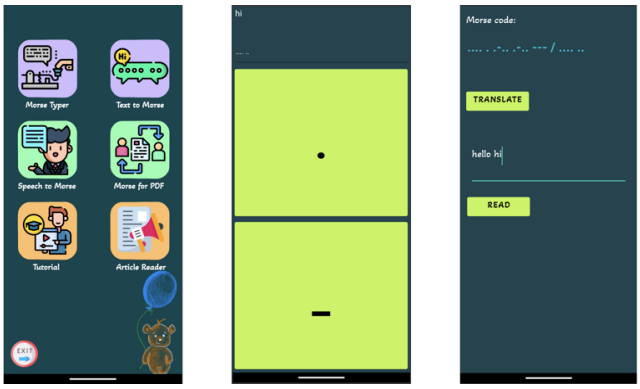


Fig. 1: a. Home Screen (left most), b. Morse Typing Window (Middle), c. Text-to-Morse Window (Right Most)

The Morse Typing Window, shown in Figure 1.b, is designed to facilitate Morse code input through a split-screen interface, whereby the top portion denotes “dots” (.) and the bottom portion denotes “dashes” (_). The working process of this window is mentioned in Figure 2. It initiates its

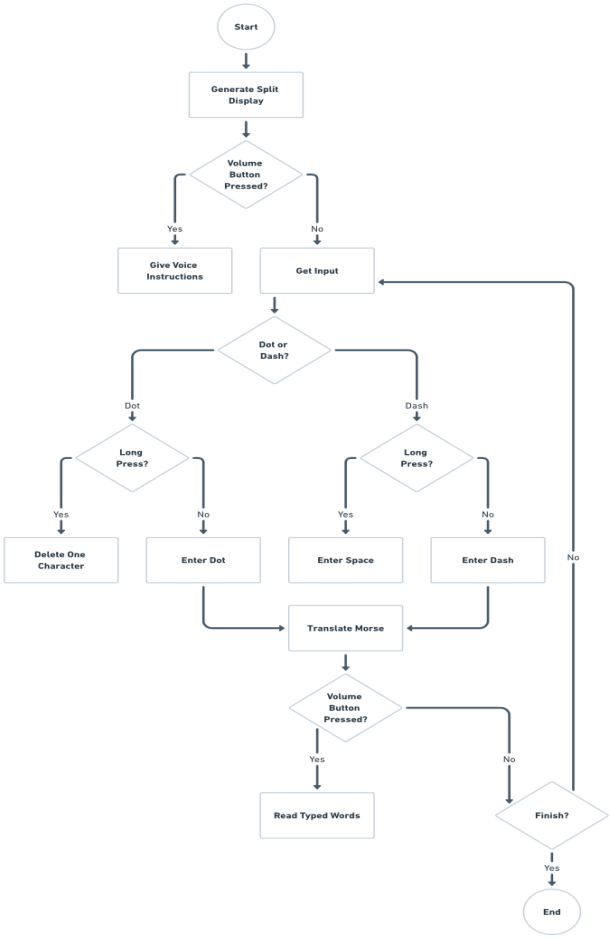


Fig. 2: The Working Process of Morse Typing Window

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processing by checking if the volume button is pressed, and upon pressing it, voice prompts guide the user through the Morse Typing Window. If the volume button is not pressed, it waits for user input. Later, the application differentiates short presses from long ones in order to detect dots and dashes. If the top portion is pressed, it is interpreted as a "dot". When the top portion is long pressed, it deletes the last morse character typed. Similarly, If the bottom portion is pressed, it is interpreted as a "dash". When the bottom portion is long pressed, it enters a whitespace character. This system not only allows correct input but also facilitates real-time correction by allowing deletion and spacing whenever needed. After catching each one of the Morse inputs, dot or dash, the application translates it into the corresponding character and then keeps listening to other inputs. If, after translation, the volume button is clicked again, it speaks out the word typed out so far, thus allowing users to cross-check their progress. This continues until the user chooses to finish, at which time the session of the Morse Typing Window closes.

B. *Text-to-Morse*

The functionality of Text-to-Morse, mentioned in Figure 1.c, is one of the most essential in ensuring increased communication for visually and hearing-impaired students. The application provides an easy interface for a user to enter texts, which then get converted into Morse code. Such conversion will definitely ensure better communication among peers and between educators and students, thus promoting a friendly educational environment. To further support understanding, the Morse code is vibrated to the students, hence creating an avenue where they can feel the messages being put across. Students are thus able to hold meaningful dialogues, consequently enhancing their learning experience and fostering a spirit of cooperation among all stakeholders involved in the learning process.

The flow of the Text-to-Morse process, as in Figure 3, beings by checking if the volume button has been pressed for voice instructions. If that is not the case, it waits for text entry. Once the text is entered, it gives the user two possibilities: "Translate" to translate the text into Morse code and "Read" to listen to it aloud.

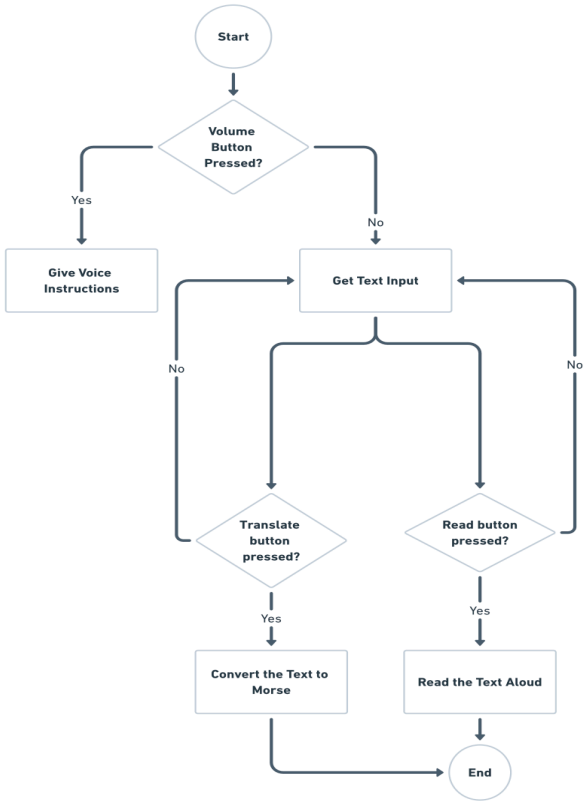


Fig. 3: The Working Process of Text-to-Morse

C. *Speech-to-Morse*

Another functionality in this application is Speech-to-Morse, which will add to the realms of accessibility, making communication easier. This interface, shown in Figure 4.a, includes a "Speak" button, which, upon being clicked, activates the microphone that picks up words spoken across it as shown in Figure 4.b. Detected speech is converted to text and then translated into Morse code by the application. It also allows the user to make the learning more interactive by pushing the "Read Morse Code" button and providing two vibrations representing the Morse code sequence of the spoken text. In this way, students with visual and auditory impairments will feel the speech, which shall increase the understanding and engagement in their minds.

Besides that, there is a "Save" button, which saves the detected speech in PDF format for future reference. Therefore, this will be one of the most taken-into-consideration benefits in educational settings because it allows teachers to instruct or explain something by only speaking, while the application translates it and informs the student both with tactile vibrations and saved records.

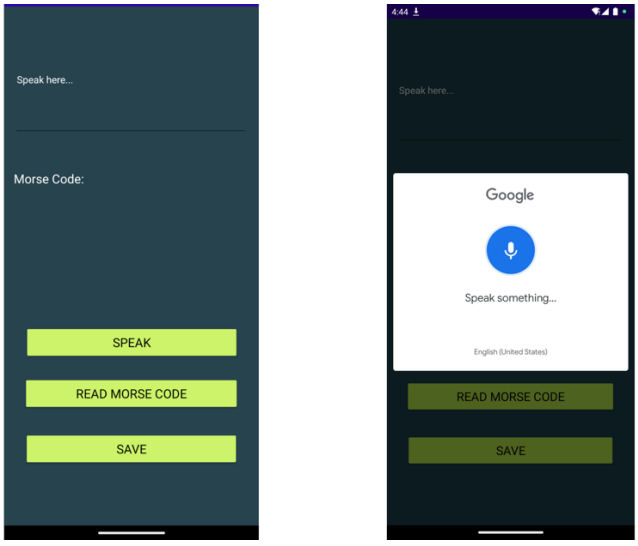


Fig. 4: a. Speech-to-Morse Window (Left) b. Accessing Mic for Speech Input (Right)

D. *Morse for PDF*

The Morse for PDF feature is designed to allow students who have both vision and hearing impairments to translate lecture materials into Morse code for learning.

When this window is opened, the user is prompted to open a PDF document to be translated into Morse code, as in Figure 5.a and Figure 5.b. Once the PDF is opened, the user

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will click “Vibrate Morse Code” which will provide them the content as Morse code vibrations. This feature would provide the users with the capability of interacting with their study materials through tactile input, hence making learning more accessible and independent.

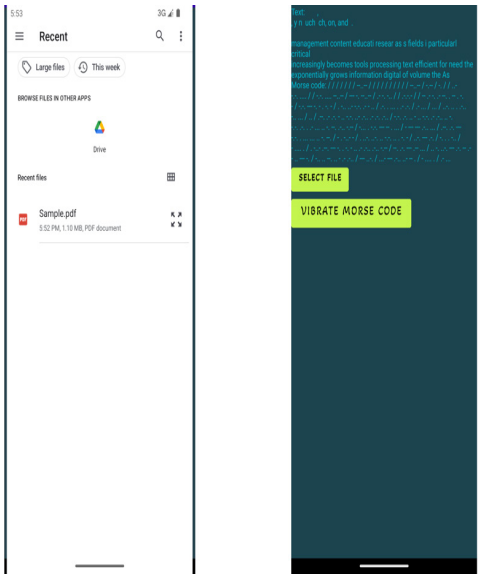


Fig 5: a. Getting PDF Input (Left) b. PDF Content into Morse (Right)

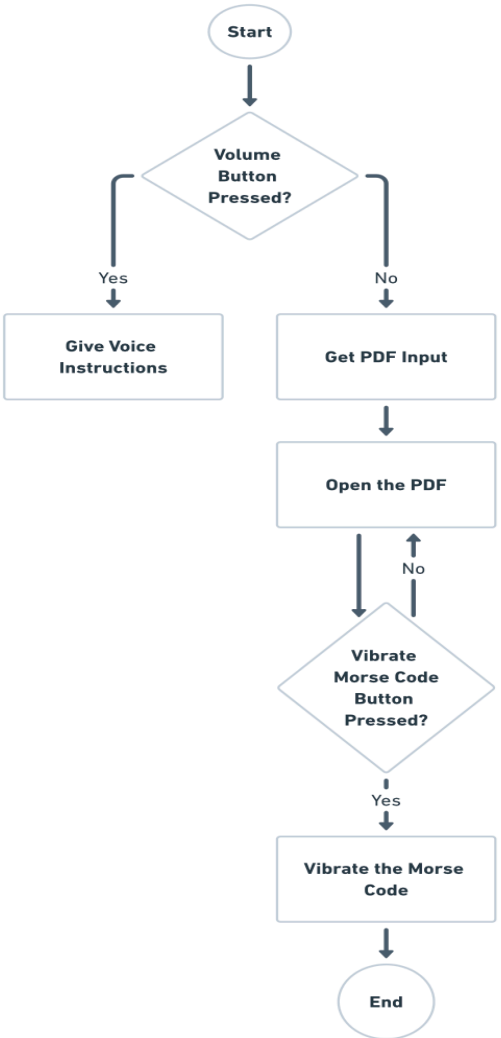


Fig. 6: Working Process of Morse for PDF

E. *Tutorial Window*

The Tutorial window, as shown in Figure 7, represents an interactive Morse learning experience with both voice and Morse instructions that take users interactively through a fun game. This allows the generation of prompts to type specific characters or numbers in Morse. This increases their acquaintance with Morse input. In this way, it is fun and easy to teach the Morse code in practice to a student targeting effective practice. This integration of audio and tactile instructions ensures that indeed users can also follow through, therefore being an effective tool in education for dual-disabled students.

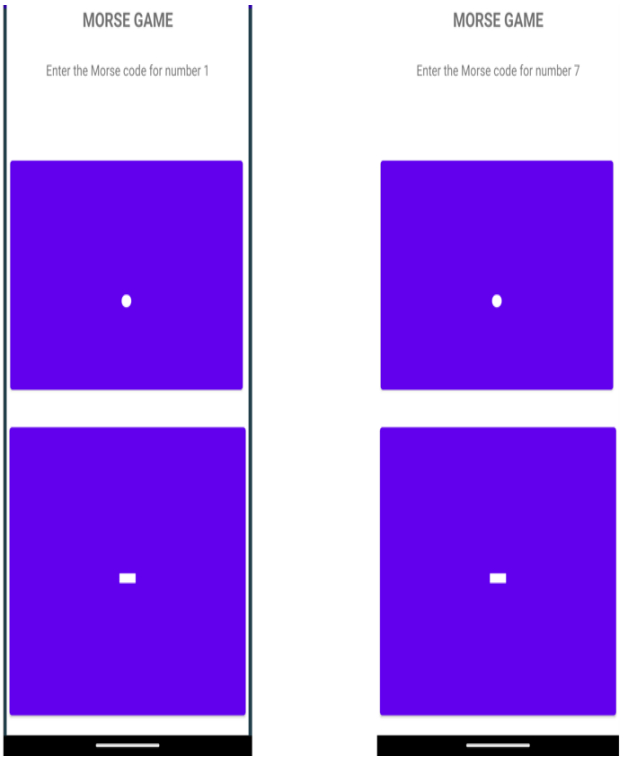


Fig. 7: Tutorial Windows

F. *Article Reader*

Article Reader aggregates articles from the internet into one place and gives a list available to be read by the user (Figure 8.a). When an article selected, it displays in both text and Morse versions to let users choose how they would like to engage with the content as seen in Figure 8.b. Clicking the Read button reads the article aloud for accessibility by visually impaired users. The selection of the Vibrate button makes the article vibrate in Morse code if clicked on, allowing dual-disabled users to feel an article by touch. Accessibility of information therefore improves, being enhanced through such adaptations of content to diverse accessibility needs.

All of the features listed above together strive to reach a point where dual-disabled students can comfortably use and access their education. Besides its pedagogical value, the application is also used as an effective communication tool because through the software, dual-disabled students who cannot write or use their vocal cords can communicate efficiently with other students and educators in Morse code.

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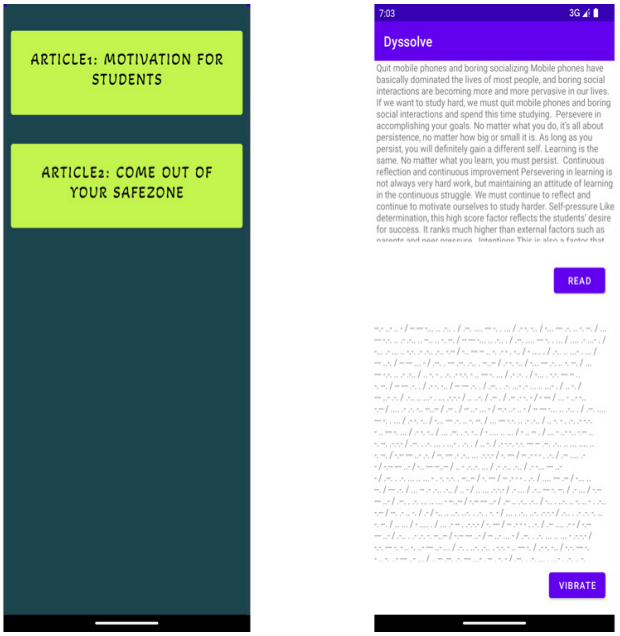


Fig. 8: a. List of Articles (Left), b. Article Content as Text and as Morse (Right)

IV. Testing Results and Evaluation

This Morse Code application was tested in a real school environment with visually impaired and dual-disabled students, in both visual and auditory impairments (refer Picture 9). Educators also participated in testing the application for its appropriateness within educational settings during schooling. During the testing, students seemed to show great interest in the application in itself, adapted to its features without any greater problems, and benefited from its dual-mode feedback of text and vibration.

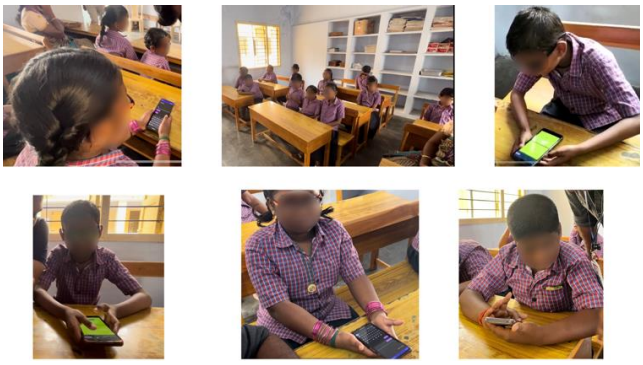


Fig. 9: First Test Campaign at School

Indeed, encouraging initial feedback led to further tests with a larger user base of 122 users, consisting of 87 visually impaired and 35 users with dual disabilities. During this enlarged test phase, users reacted against questions related to satisfaction showing a return breakdown of 82% positive, 10% neutral, and 8% negative claims (as plotted in Figure 10).

The application was installed on the participants' devices and was monitored for seven months, starting from February 2024 to August 2024 (as plotted in Figure 11). The usages of the application in this period went on increasing significantly month after month, showing that it was of continued value and adaptive in the daily life of the users.

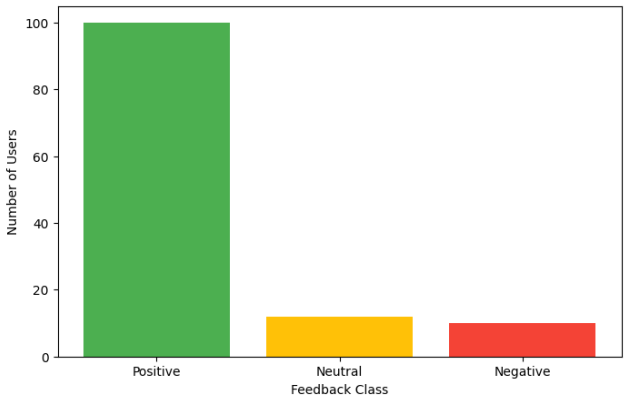


Fig. 10: Feedback Distribution

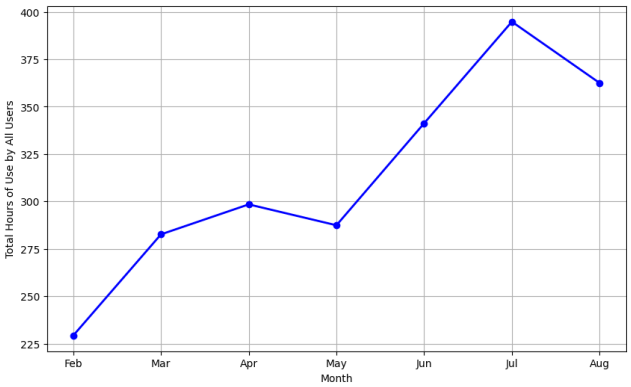


Fig. 11: Application Usage Over 7 Months

V. Conclusion

This paper proposed the implementation of a Morse code-based application that could empower people with visual and dual disabilities in the contexts of education and communication. Real-time haptic feedback, split-screen Morse typing, and NLP-driven text suggestions within the application serve to empower users in learning and communicating effectively. It also went live in many real-school environments, coming out with overwhelmingly positive feedback while showing huge potential to improve accessibility and learning for students with disabilities. The general illustration of this application is how technology bridges gaps in various ways of accessibility that students with disabilities need, and the promotion of inclusion in educational settings. In the continuous development of this platform, further improvements will focus on expanding features and increasing usability to improve learning for an even wider audience.

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